Vector field

Defn: A vector field $\vec{F}$ on $D \subseteq \mathbb{R}^n$ assigns to each $x = (x_1, x_2, \ldots, x_n)$ a vector $\vec{F}(x)$ in $\mathbb{R}^n$.

$$\mathbb{R}^n \xrightarrow{\vec{F}} \mathbb{R}^n$$

For $\mathbb{R}^2$ and $\mathbb{R}^3$, we can use the notation:

* $\vec{F} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$$(x,y) \longmapsto \vec{F}(x,y) = P(x,y)\,\vec{i} + Q(x,y)\,\vec{j} = \langle P(x,y), Q(x,y) \rangle$$

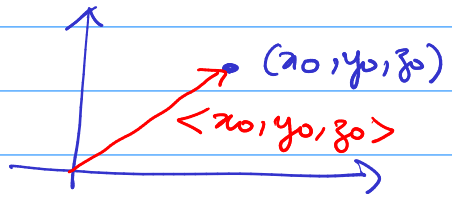
* $\vec{F} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$$(x,y,z) \longmapsto \vec{F}(x,y,z) = \langle P(x,y,z), Q(x,y,z), R(x,y,z) \rangle$$

Remark: $(x_0, y_0, z_0) \in \mathbb{R}^n$ as a point.

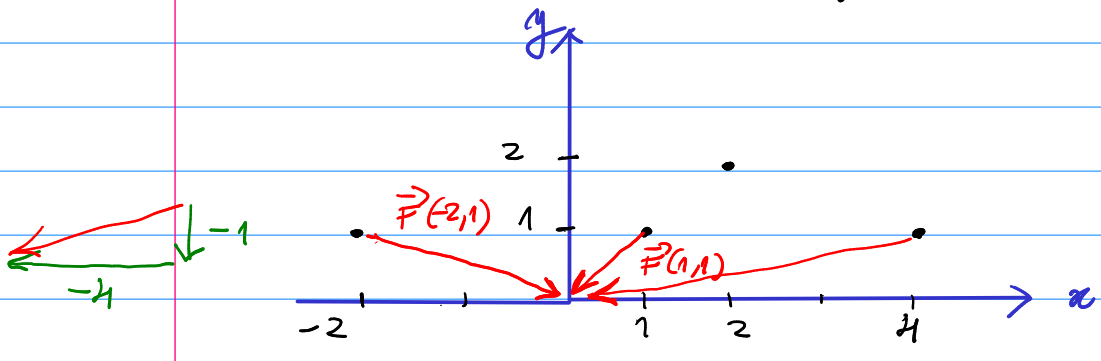
$\Rightarrow$ gives a vector $\langle x_0, y_0, z_0 \rangle$

(x,y)
$\langle x,y \rangle$



Examples: (a) $\vec{F} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$$(x,y) \longmapsto \langle -x, -y \rangle = -\langle x,y \rangle$$



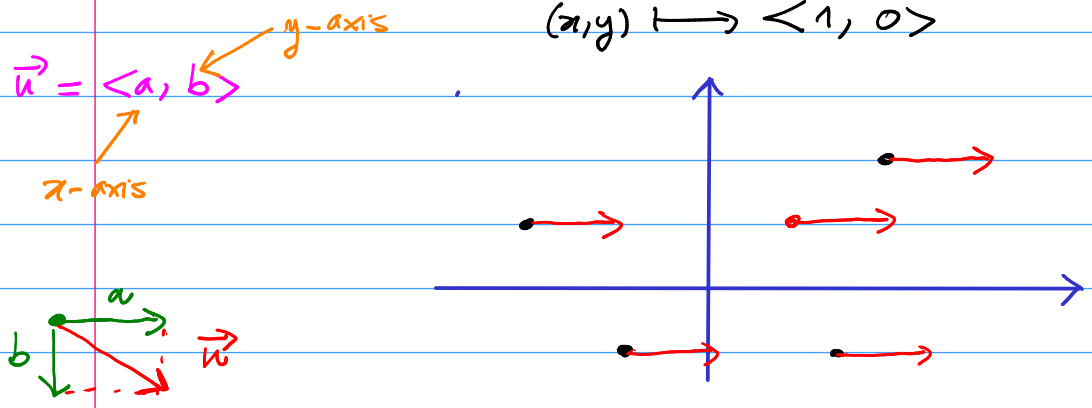
$\vec{F}(1,1) = \langle -1, -1 \rangle$

$\vec{F}(4,1) = \langle -4, -1 \rangle$

$\vec{F}(-2,1) = \langle 2, -1 \rangle$

(b) $\vec{F}: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$(x,y) \longmapsto \langle 1, 0 \rangle$



Assumption: $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ exist

Defn: For a function (multivariate) $f: \mathbb{R}^n \longrightarrow \mathbb{R}$, the gradient ∇f is a vector field. It is called "gradient vector field".

- $f: \mathbb{R}^2 \longrightarrow \mathbb{R}$

$$\Downarrow$$

$$\nabla f: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$(a,b) \longmapsto \left\langle \frac{\partial f}{\partial x}(a,b), \frac{\partial f}{\partial y}(a,b) \right\rangle$$

$$\nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j}$$

- $f: \mathbb{R}^3 \longrightarrow \mathbb{R}$

$$\Rightarrow \nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}$$

Defn: A vector field $\vec{F}$ is conservative if there is a function $f: \mathbb{R}^n \longrightarrow \mathbb{R}$ such that $\vec{F} = \nabla f$.

Example: $\vec{F}(x,y) = \langle 2x, -2y \rangle$ ($P(x,y)$ is the $2x$ component, $Q(x,y)$ is the $-2y$ component)

$\Rightarrow$ If $f(x,y) = x^2 - y^2$, then $\vec{F} = \nabla f$.

$\vec{F} = f_x \vec{i} + f_y \vec{j}$

$2x = \frac{\partial f}{\partial x} = f_x$ and $-2y = \frac{\partial f}{\partial y} = f_y$

$\vec{F}(x,y) = \left\langle \frac{\partial f}{\partial x}(x,y), \frac{\partial f}{\partial y}(x,y) \right\rangle$